DELHI PUBLIC SCHOOL · DEPARTMENT OF PHYSICS

MID-TERM EXAMINATION (CLASS XI)

Subject: Physics (Theory) Maximum Marks: 20 | Time: 1 hr

Q1 [4 Marks]

State Newton's Second Law of Motion and derive its mathematical formula (F = ma).

Q2 [6 Marks]

A 1500 kg automobile accelerates uniformly from rest to 20 m/s in 10 s. Calculate:

(a) Acceleration of the car,

(b) Net horizontal force acting on it.

Q3 [5 Marks]

Define Work. Calculate the work done when a force of 50 N moves a crate through a distance of 6 m across a frictionles

Q4 [5 Marks]

Distinguish clearly between Kinetic Energy and Potential Energy. Give one practical example illustrating each type.